

BIO DATA



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Google Scholar link: <https://scholar.google.co.in/citations?hl=en&user=S2jByOUAAAAJ>

1. Educational Qualification:

Degree	Branch / Specialization	Institute/University	Year of Passing
B.E.	Electrical and Electronics Engineering	Mepco Schlenk Engineering College, Sivakasi	2012
M.E.	Power Systems Engineering	College of Engineering Guindy, Anna university Chennai	2014
Ph.D.	Power System Planning	Anna University	2024

2. Professional/ Industry Experience: (in chronological order)

S.No	Designation	Institution/ Organization	Period	
			From	To
1	Associate Professor - I	Ramco Institute of Technology, Rajapalayam	06.03.2024	Till Date
2	Assistant Professor (Senior Grade)	Ramco Institute of Technology, Rajapalayam.	01.01.2023	05.03.2024
3	Assistant Professor	Ramco Institute of Technology, Rajapalayam	01.06.2015	31.12.2022
4	Assistant Professor	M. Kumarasamy College of Engineering, Karur	16.06.2014	18.05.2015

3. Research Interest: Power Systems, Soft Computing Techniques, Internet of Things, Renewable Energy Systems, Energy Storage Systems.
(Anna University Recognized Research Supervisor)

Updated as on **01.04.2026**

4 Publications

4.1 JOURNAL PUBLICATIONS

- 1) Rajasekaran, S., **Arun Kumar, A.**, Thangavel, C., & Arunachalam, K. P. (2025). Artificial intelligence enabled internet of things for modeling a smart power distribution system. Journal of the Chinese Institute of Engineers, 1–19. <https://doi.org/10.1080/02533839.2025.2578199>. (SCIE indexed)
- 2) Rajalakshmi, J., Jeyanthi, S., Bhuvanesh, A, **Arun Kumar A.**, (2024). Analysis of Solar Photovoltaic Integration and Plug-in Electric Vehicle Charging Systems with a Bidirectional Buck-Boost DC to DC Converter. Journal of The Institution of Engineers (India): Series B <https://doi.org/10.1007/s40031-024-01185-x> (Scopus Indexed)
- 3) **Arun Kumar, A.**, Geetha, M., Ramkumar, A. *et al.* Generation expansion planning incorporating the recuperation of older power plants for economic advantage. Electr Eng (2024). <https://doi.org/10.1007/s00202-024-02708-x> (SCIE indexed)
- 4) **Arun Kumar, A.** et al. 2024, ‘A novel generation expansion planning in older power plants – hybrid spotted hyena-particle swarm optimization’ Opto-Electronics Review, vol. 32, no. 3. (SCIE indexed)
- 5) **Arun Kumar, A.** ‘Empowering Remote Sensing: Ant Colony Optimization and RCNN for Precision Environmental Monitoring’ ICTACT Journal on Soft Computing, July 2024 vol. 15, no. 1. (UGC care Approved)
- 6) **Arun Kumar, A.** et al. 2024, ‘Recuperation Based Multi-objective Generation Expansion Planning for a Real-world Power System’ Journal of Intelligent & Fuzzy Systems, vol. 45, no. 4, pp. 6751 – 6766. (SCIE indexed)
- 7) **Arun Kumar Athiappan**, Manikandan Bairavan Veerayan & Kannan Subramanian 2023, ‘Multi-Objective Based Generation Expansion Planning for Utility Power System of Tamil Nadu State Considering Recuperation of Older Power Plants’, Electric Power Components and Systems, vol. 51, no. 17, pp.1992-2009 (SCIE indexed)
- 8) Jeyanthi S, **Arun Kumar A**, Muthu Rani M M , Rohini M I , Divya Jeyashree V “IoT Based Food Waste Monitoring and Controlling System”, Journal of Pharmaceutical Negative Results. Issue No.3, Volume.14 (Feb. 2023), 586–591. DOI:<https://doi.org/10.47750/pnr.2023.14.03.073>.
- 9) **Arun Kumar Athiappan**, Manikandan Bairavan Veerayan, Kannan Subramanian and Bhuvanesh A, “Application of Cognitive Optimization Algorithms to Solve Generation Expansion Planning Problem by Considering the Recuperation for Aged Thermal Plants”, NeuroQuantology; Bornova Izmir Vol. 20, Issue 8, (2022): 7875 - 7887. DOI:10.14704/nq.2022.20.8.NQ44813. (Scopus Indexed).
- 10) S. Amosedinakaran, T. A. Sunil Raj, R. Robin Jesubalan & **A. Arun Kumar**, “Scada Based Monitoring and Control of DC Motor”, Natural Volatiles & Essential oils, Vol. 8, No. 4, pp. 1587-1601, November 2021. (SCOPUS indexed)
- 11) R. Jai Ganesh, **A. Arun Kumar**, B. Devi Sri, P. Pragadeesh and N. Srivatshan, “IoT based Covid Patient health monitor during self-quarantine”, Turkish Online Journal of Qualitative Inquiry (TOJQI), Volume 12, Issue 6, July, 2021: 7750-7762.
- 12) M.Suba Sathya, J.H.Aslam Javid Khan , M.Velmurugan, **A.Arun Kumar**, Forestalling fire accident in train, International Research Journal of Engineering and Technology (IRJET), Volume: 07 Issue: 03, Mar 2020. Impact Factor: 7.34 (UGC Approved)

- 13) **Arun Kumar A**, B.V. Manikandan ,S. Kannan, A Review on Energy Storage Systems in Distributed Generation for Generation Expansion Planning Study, International Journal of Advanced Science and Technology, 29(7s), 1103 - 1112 (2020). Impact Factor: 0.11. **(Scopus indexed)**
- 14) S. Amathulla Jasila, S. Anupriya, S.R. Deve Kala, T.Priyanga Gandhi, and **Arun Kumar A**, published a paper entitled “PV Grid interconnected Renewable Energy System“, SSRG International Journal of Electrical and Electronics Engineering (SSRG - IJEEE), ICRTECITA April 2018.
- 15) P.Preetha, **Arun Kumar A** “Stability Improvement Using Unified Power Flow Controller (UPFC)” International Journal of Advanced Research In Electrical, Electronics and Instrumentation Engineering in IJAREEIE, Volume 4, Special Issue 2, April 2015.

4.2 CONFERENCE PUBLICATIONS

- 1) Praveenkumar V, Sivasubramanian M, Sridhar S and **Arun Kumar A** presented a paper titled “Smart Parking Solution for Urban” in National Conference on Advances in Science, Technology and Management (NCASTM-25) at Info Institute of Engineering, Tamilnadu on April 30, 2025.
- 2) Dhanush Balaji S, Kali Karthick P, Vishaal M and **Arun Kumar A** presented a paper titled “IoT-Based Smart and Secure Multipurpose Helmet for Two-Wheeler” in International conference on Recent Advances and innovations in science, Engineering, Technology and management (ICRAISETM-2025) at Solamalai College of Engineering, Madurai, Tamilnadu on April 25 & 26, 2025.
- 3) **Arun Kumar A**, Rajakarunakaran S and Ganesan L, presented a paper titled “Achieving program outcomes in engineering education: Strategies for curriculum integration and assessment refinement”, in the 2nd International Conference on Educating the Post Millennials (ICEM)’24 at Saveetha Teaching Learning Centre (STLC) at Saveetha Engineering College, during May 17, 2024.
- 4) Arunkumar S, Gayathree P G, M.Karpagaraj and **Arun Kumar A**, presented a paper titled “Enhancement of Resilience using Smart Fault Detection and Localization in Electrical System”, in the International Conference on Exploring AI, IOT, Science and Technology (ICEAIST-24) at M.A.M School of Engineering, Trichy, during April 26 & 27, 2024.
- 5) Jeyanthi S, Seenu RA, Hariharasudhan M, Rishi Raghul, **Arun Kumar A**, presented a paper titled “Digital Multimeter”, in the International Conference on Smart Engineering for Renewable Energy Technologies (ICSERET 2023), Ramco Institute of Technology, during March 25, 2023.
- 6) **Arun Kumar A**, Muthu Ramya R, Jeyapriyadharshini M, Jeyanthi S, presented a paper titled “Warning Sign Cataloging Using Neural Network”, in the International Conference on Smart Engineering for Renewable Energy Technologies (ICSERET 2023), Ramco Institute of Technology, during March 25, 2023.
- 7) **Arun Kumar A** , M. M. Muthu Rani, M.I. Rohini, V. Divya Jeyashree, S. Jeyanthi presented a paper titled “Smart Garbage Detection System”, in the International Conference on Smart Technologies and Applications (ICSTA 2022) organized by Ramco Institute of Technology, Rajapalayam during March 11 - 12, 2022.
- 8) R. Maha Guru Backiyam, T. Munieshwari, P. Sangeetha, **A. Arun Kumar**, S. Jeyanthi, presented a paper entitled "Improvement of efficiency of solar cells using nano coating" in the National Web Conference on Challenges and Innovation in Engineering & Technology

2021, during 19 - 20 March 2021.

4.3 Book/Book chapter publications

1. **Title of the Book:** Basics of Electrical and Electronics Engineering,
Publisher: Anuradha Publications
Year: 2025
ISBN: 978-93-91677-82-4
2. **Title of the Book:** Changing Face of Agriculture, Agri- Business, Agriculture Marketing and Organic Farming- Innovation and Strategies
Chapter Title: Challenges of Municipal Solid Waste Management: An Empirical Study
Publisher: Iterative International Publishers
Volume: 01
ISBN: 978-81-959356-2-8

5. Funded Projects

S.No.	Title of the project	Funding Agency	Duration	Amount	Status
1.	LED Powered Endo Illuminator for Vitreous Surgery	Ministry of Micro, Small & Medium Enterprises (MSME)	1 year	Rs. 14,50,000/-	On-going

6. Ph.D. guidance

Sl.No	Scholar name	University	Topic	Status
NIL				

7. Short term Courses / Seminars /FDP/Workshop/Conference Attended:

Sl. No	Title of the event	Organized by	Date	
			From	To
1.	ATAL FDP on The Future of Pedagogy: ChatGPT and AI Tools as Teaching Assistants	AVS Engineering College	05.01.2026	10.01.2026
2.	FDP on Competency Based Engineering Education	Ramco Institute of Technology	10.02.2025	15.02.2025
3.	ATAL FDP on Climatic Crisis and Mitigation through Sustainable Development of Energy and Environment Systems	Sri Krishna College of Engineering and Technology	16.12.2024	21.12.2024
4.	FDP on Teaching And Learning in Engineering (TALE)	NPTEL	01.01.2025	28.02.2025
5.	Extramural Lecture Series (ELS)- Mapping the CDIO Syllabus for Sustainability	Ramco Institute of Technology	20.12.2024	-
6.	Seminar on Being 'Industry Ready' with System-Modelling Skills	MathWorks	01.06.2024	-

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Sl. No	Title of the event	Organized by	Date	
			From	To
7.	Design of Electrical Vehicles using MATLAB Simulink	Ramco Institute of Technology	24.07.2023	26.07.2023
8.	Accreditation and Academic Excellence	Vellore Institute of Technology, Chennai	21.07.2023	-
9.	Extramural Lecture Series (ELS)-ELS001 - Integrated Curriculum Development Methodology for Engineering using CDIO Framework: A Case Study	Ramco Institute of Technology	16.11.2023	-
10.	FDP on “NBA Accreditation and Teaching and learning in Engineering”	NPTEL	Jan 2023	Apr 2023
11.	Train the Trainer “Powering IoT using Arduino / RaspberryPi”	SkillsDA	19.09.2022	23.09.2022
12.	TinyML for scientific use cases	Ramco Institute of Technology	25.07.2022	-
13.	Skill Development Training Program on “Internet of Things – Phase II”	ISTE & BNY Mellon.	25.06.2022	11.08.2022
14.	Online Elementary FDP on "Outcome Based Learning"	Viveknand Education SocietyBusinessSchool.	11.01.2022	15.01.2022
15.	Three day workshop on Examination Reforms	AICTE	27.12.2021	29.12.2021
16.	Online FDP on “UniversalHuman Value”	AICTE	06.12.2021	10.12.2021
17.	ATAL FDP on Soft Computing Techniques	Dr. S. & S. S. Ghandhy Government Engineering College, Gujarat	02.11.2020	06.11.2020
18.	2 Days Online Certified Workshop on Pvsyst	KWATT Solutions Private Limited, Bombay	28.10.2020	29.10.2020
19.	FDP on Research Areas in Energy Audit & Conservation	VIT, Vellore	24.08.2020	26.08.2020
20.	Research Opportunities in Electrical Engineering and its Applications	P.S.R. Engineering College Sivakasi	18.05.2020	23.05.2020
21.	One Day FDP on “Renewable Energy Systems Laboratory-Hands on Training"	Velammal College of Engineering and Technology, Madurai	16.12.2019	-
22.	FDP on “Intellectual Property Rights”	Ramco Institute of Technology Sponsored by Anna University	02.12.2019	07.12.2019

Sl. No	Title of the event	Organized by	Date	
			From	To
23.	Three-Day FDP on Team Building	Ramco Institute of Technology in Association with ICT Academy of Tamil Nadu	18.11.2019	20.11.2019
24.	Application of Advanced Software Tools For Smart Power System	SSN College of Engineering, Chennai	08.08.2019	09.08.2019
25.	Energy Conclave 2018	Department of EEE, Thiagarajar College of Engineering, Madurai	17.12.2018	19.12.2018
26.	Mathematical Optimization Methods for Power Systems	Centre for Continuing Education, IISc, Bengaluru	18.06.2018	22.06.2018
27.	Intelligent Electrical Power Grids (IEPG)	NIT, Tiruchirappalli	27.11.2017	01.12.2017
28.	FDP on Power System Operation and Control	College of Engineering Guindy, Chennai	23.11.2016	29.11.2016
29.	Workshop on “Operation and Control of Wind-driven Generators”	NIT, Tiruchirappalli	15.04.2016	16.04.2016
30.	National Workshop On “Electric PowerQuality”	NIT, Calicut	11.03.2016	12.03.2016
31.	STTP on Power Electronics: System and Control	SVNIT, Surat	14.12.2015	18.12.2015
32.	Recent Trends in Power System Protection	NIT, Warangal	27.11.2014	29.11.2014

8. Short term Courses / Seminars /FDP/Workshop/Conference organized:

Sl.No	Title of the event	Organized by	Date	
			From	To
1.	Industry-Led Guest Lecture on Home Automation	Ramco Institute of Technology, Rajapalayam	04.03.2025	-
2.	STTP on “Fundamentals of Electrical and Electronics”	Ramco Institute of Technology, Rajapalayam	10.07.2024, 15.07.2024	16.07.2024
3.	Five Days Value Added Course on “Internet of Things”	Ramco Institute of Technology, Rajapalayam	19.06.2023	23.06.2023
4.	Workshop on “Applications of Tinkercad to Electrical Engineering”	Ramco Institute of Technology, Rajapalayam	17.11.2022	-
5.	Workshop on “Tinkercad for Tomorrow’s Technologists”	Ramco Institute of Technology, Rajapalayam	27.04.2023	-

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Sl.No	Title of the event	Organized by	Date	
			From	To
6.	Value Added Course on “Embedded Controllers and its Applications in Electrical Engineering”	Ramco Institute of Technology, Rajapalayam	10.08.2022	13.08.2022
7.	3-Day Online FDP on Bio Inspired Optimization Techniques for Electrical Engineering Applications	Ramco Institute of Technology, Rajapalayam	21.02.2022	23.02.2022
8.	Fun with ElectricCircuits	Ramco Institute of Technology, Rajapalayam	11.05.2020	13.05.2020
9.	Five day finishing school programme on Hands on Training on Arduino Programmingand hardware	Ramco Institute of Technology, Rajapalayam	22.04.2019	26.04.2019
10.	National level Two- Day workshop on Arduino Programmingand Hardware	Ramco Institute of Technology, Rajapalayam	11.01.2019	12.01.2019
11.	One day workshop on “Arduino” in Techyuga 2018	Ramco Institute of Technology, Rajapalayam	31.08.2018	-

9. Online Courses

Sl.No	Course title	University	Date
1.	OBE and Accreditation	IISc Bangalore	July-Oct 2025
2.	Teaching And Learning in Engineering (TALE)	Indian Institute of Science Bangalore	Jan-Feb 2025
3.	Introduction to Graph Theory	University of California San Diego	19.10.2024
4.	Mathematics for Machine Learning: Multivariate Calculus	Imperial College London	03.05.2024
5.	Differential Equations for Engineers	The Hong Kong University of Science and Technology	18.06.2023
6.	NPTEL Course on “NBA Accreditation and Teaching and learning in Engineering”	NPTEL	Jan-Apr 2023
7.	Getting Started with Deep Learning	NVIDIA	23.11.2022
8.	Mathematics for Machine Learning: Linear Algebra	Imperial College London	31.03.2022
9.	Introduction to Programming with MATLAB	Coursera	09.09.2021
10.	Solar Energy: Integration of Photovoltaic Systems in Microgrids	edX	28.10.2020
11.	Linear Circuits 1: DC Analysis	Coursera	20.09.2020
12.	Operations Research: an Active Learning Approach	edX	03.09.2020

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Sl.No	Course title	University	Date
13.	Transfer functions and Laplace transforms	edX	01.09.2020
14.	Programming for Everybody (Getting Started with Python)	Coursera	25.06.2020
15.	Electric Power Systems	Coursera	25.06.2020
16.	Vector Calculus for Engineers	Coursera	12.06.2020
17.	Introduction to Differential Equations	MITx	26.12.2019
18.	Circuits and Electronics 2: Amplification, Speed, and Delay	MITx	31.05.2019
19.	Calculus 1C: Coordinate Systems & Infinite Series	MITx	12.05.2019
20.	Differential Equations: Fourier Series and Partial Differential Equations	MITx	21.12.2018
21.	Calculus 1B: Integration	MITx	09.03.2018
22.	Calculus 1A: Differentiation	MITx	10.09.2017
23.	Solar Energy	Delft University, Netherland	06.06.2017
24.	Smart Grid	IEEEEx	23.12.2016
25.	Circuit and Electronics 1: Basic Circuit Analysis	MITx	09.05.2016

10. Patents filed/awarded: -

Sl. No.	Title	Application No.	Filed Date	Status	Inventor(s)
1.	Multi Terminal Corrosion Free Service Line Clamp	202141004396	01.02.2021	Granted	Mrs. S. Jeyanthi Mr. A. Arun Kumar Dr. S. Kannan Dr. G. Kanthimathi Dr. O. Senthil Kumar
2.	Electronic Control for Electric Vehicle	202241017752	28.03.2022	Granted	Mrs. S. Jeyanthi Mr. A. Arun Kumar Mr. R. Gajendra Mr. K. Vignesh Mr. E. Athinam
3.	Safety-Enhanced Ceiling Fan System for Suicide Deterrence	202441062081	30.08.2024	Published	Mrs. S. Jeyanthi Ms. G. Dhana Bharathi Lakshmi Ms. P.G. Gayathree Ms. B. Jeyapadma Dr. A. Arun Kumar
4.	Automatic System for Pandemic Preparedness in Smart Hospital	202441063028	30.08.2024	Published	Dr. A. Arun Kumar Ms. M. Lakshmi Mrs. S. Jeyanthi

11. Copyright filed/awarded:

Sl.NO.	Title of Work	Registration Number / Diary Number	Status	Name of the Author(s)
1.	Handbook on Renewable Energy Systems Laboratory	L-122561/2023	Granted	Mr. A. Arun Kumar, Mrs. S. Jeyanthi
2.	Laboratory Manual on Grid Integrating Techniques and Challenges	27864/2024-CO/L	Granted	Mrs. S. Jeyanthi, Dr. A. Arun Kumar
3.	Laboratory Manual on Smart System Automation	38680/2024-CO/L	Granted	Dr. A. Arun Kumar Mrs. S. Jeyanthi

12. Membership in Professional societies : ISTE and IEANG**13. Awards, Recognition & Achievements**

- Ranked as the **Top 1% Achiever** in **all India level** in the Prestigious **NPTEL** Course on ‘OBE and Accreditation’
- Performing as **Top Performing Mentor** for the Prestigious **NPTEL** Course on ‘Introduction to Internet of Things’.
- Received **Research Supervisor Recognition** from Anna University, Chennai in the year of 2024 with University Recognition Number: 4430021.
- Ranked as the **Top 1% Achiever** in **all India level** in the Prestigious **NPTEL** Course on ‘Teaching and Learning in Engineering (TALE)’
- **Certified with Distinction** under the prestigious **IUCEE** International Educator Certification Program in Fall 2023
- Ranked as the **Top 1% Achiever** in **all India level** in the Prestigious **NPTEL** Course on ‘NBA Accreditation and Teaching and Learning in Engineering (NATE)’
- Received **best class adviser** award for the academic year 2019-20 (2017-21 batch)
- **Produced 100% result** in EE6004 – Flexible AC Transmission System in academic year 2017 – 18 Odd Semester.
- **Produced 100% result** in EE6603 – Power System Operation and Control in academic year 2015 – 16 Even Semester.
- **Produced 100% result** in EE3020 – Smart System Automation in academic year 2023-24 Even Semester and 2024-25 Even Semester.
- Received **Best Library User** Certificate for the Academic Year 2021-2022
- Received **Best IEEE Student Branch award** for the year 2020.
- Recognized as a **NAAC Mock Auditor** for SSM College of Engineering, Dindugal
- Recognized as a department level **mentor for NBA Accreditation preparation** for Polytechnic colleges such as Angel Polytechnic College, Tenkasi and Sankar Polytechnic College, Tirunelveli.

Any others (Board of studies/Governing Council/Staff Selection/ Reviewer/other duties)

- Controller of Examinations from 01.10.2022
- Mock Auditor – NBA and NAAC
- Acted as Naan Mudhalvan Single Point of Contact (SPoC) for the academic year 2022-23
- In-Charge for Examination Control office since 01.01.2018
- Member in Internal Quality Assurance Cell (IQAC) for NAAC Accreditation
- PAQIC Member for Department of Mechanical Engineering
- Reviewer for the Journals: e-Prime - Advances in Electrical Engineering, Electronics and Energy, Energy Reports, Environment, Development and Sustainability.